

CLOUD COMPUTING ASSIGNMENT

Implementation of IAM Policies, Secure Storage and Data Encryption on AWS

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DOMAIN	CLOUD COMPUTING

Assignment Report

Cloud Platform	Amazon Web Services (AWS)
Primary Services	AWS IAM and Amazon S3
S3 Bucket	student-cloud-security-2026
Report Type	Configured Security Policies and Implementation Report

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1. Introduction

Cloud platforms provide scalable storage and computing resources, but cloud resources must be properly secured to prevent unauthorized access and data exposure. This practical demonstrates three fundamental cloud-security controls on Amazon Web Services (AWS): Identity and Access Management (IAM), secure object storage using Amazon S3, and encryption of data at rest.

The implementation uses an Amazon S3 bucket named student-cloud-security-2026 for storing sample objects. An IAM user named student-user is configured with a custom read-only policy for controlled access to the S3 resources. S3 server-side encryption with Amazon S3 managed keys (SSE-S3) is enabled for stored data.

2. Aim

To implement IAM policies, secure cloud storage, and data encryption on Amazon Web Services and document the resulting security configuration.

3. Objectives

- Create and configure an Amazon S3 bucket for secure cloud storage.
- Upload sample files to the S3 bucket.
- Create an IAM user and apply a least-privilege style S3 access policy.
- Restrict public access to the storage resource.
- Enable and verify server-side encryption using SSE-S3.
- Collect screenshots as evidence and document the implementation.

4. AWS Services Used

Service	Purpose	Security Role
Amazon S3	Object storage for sample files	Private storage, access control and encryption
AWS IAM	Identity and permission management	User and policy-based authorization

5. Security Architecture

The implemented security flow is: AWS account → IAM identity → IAM policy → Amazon S3 bucket → encrypted objects. Public access is intended to remain blocked, while authorized IAM permissions control access to the stored objects.

AWS Account	IAM User	IAM Policy	S3 Bucket	Encrypted Objects
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6. Implementation Procedure

6.1 AWS Console Access

Access the AWS Management Console and open the services required for the practical.

6.2 S3 Bucket Setup

Create the S3 bucket student-cloud-security-2026 in the selected AWS Region.

6.3 Object Upload

Upload sample objects to the S3 bucket and verify that they appear in the Objects tab.

6.4 IAM Configuration

Create the student-user IAM identity and attach the custom S3ReadOnlyPolicy.

6.5 Storage Security

Keep the S3 bucket private and verify that public access is blocked.

6.6 Encryption

Verify default server-side encryption using Amazon S3 managed keys (SSE-S3).

7. IAM Policy Configuration

An IAM user named student-user was configured for the practical. A custom policy named S3ReadOnlyPolicy was attached directly to the user. The policy is intended to permit read/list operations on the selected S3 bucket without granting broad administrative permissions.

Example policy structure used for the bucket:

```
{
  "Version": "2012-10-17",
  "Statement": [
```

```
{
  "Sid": "ReadOnlyS3Bucket",
  "Effect": "Allow",
  "Action": [
    "s3:GetObject",
    "s3:ListBucket"
  ],
  "Resource": [
    "arn:aws:s3:::student-cloud-security-2026",
    "arn:aws:s3:::student-cloud-security-2026/*"
  ]
}
```

8. Secure S3 Storage Configuration

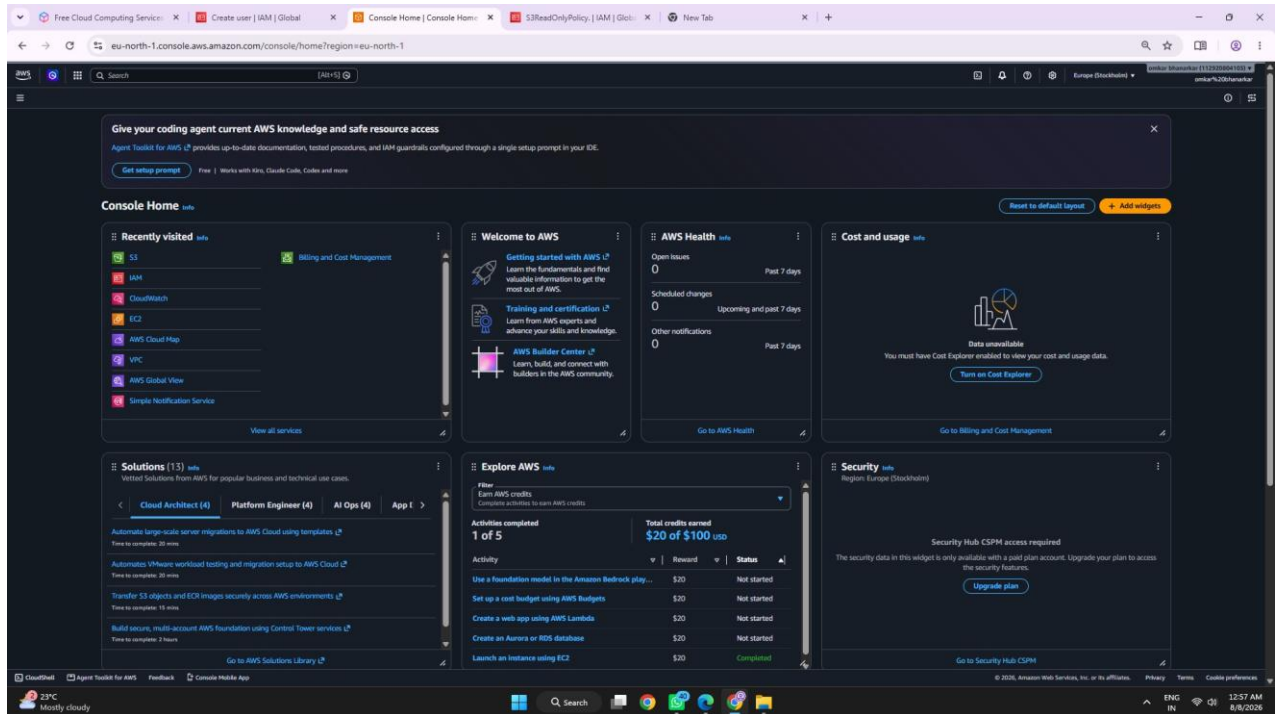
The Amazon S3 bucket is used as the secure storage layer. Sample objects were uploaded successfully. For secure storage, public access should remain blocked so that objects are not anonymously accessible. Access is instead controlled through AWS permissions.

9. Data Encryption Configuration

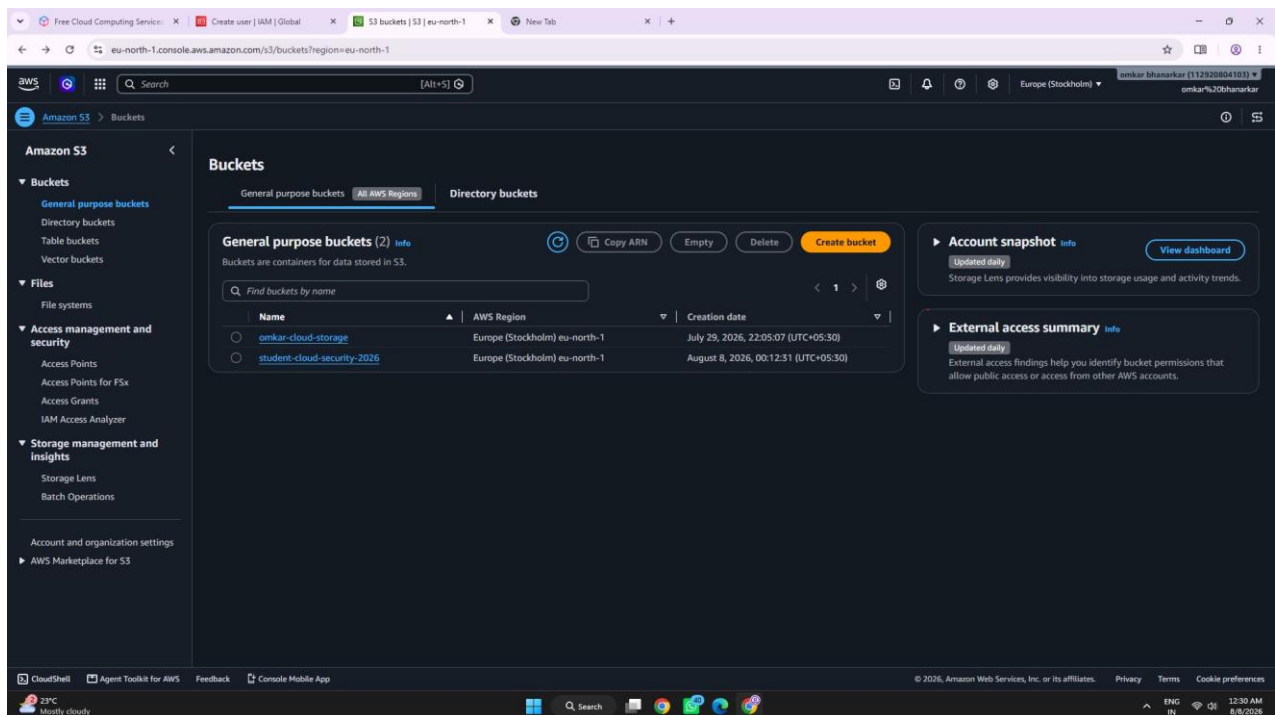
Default server-side encryption is enabled for the bucket. The captured AWS console evidence shows Server-side encryption with Amazon S3 managed keys (SSE-S3), with the S3 Bucket Key enabled. This provides encryption at rest for new objects stored in the bucket.

10. Evidence and Screenshots

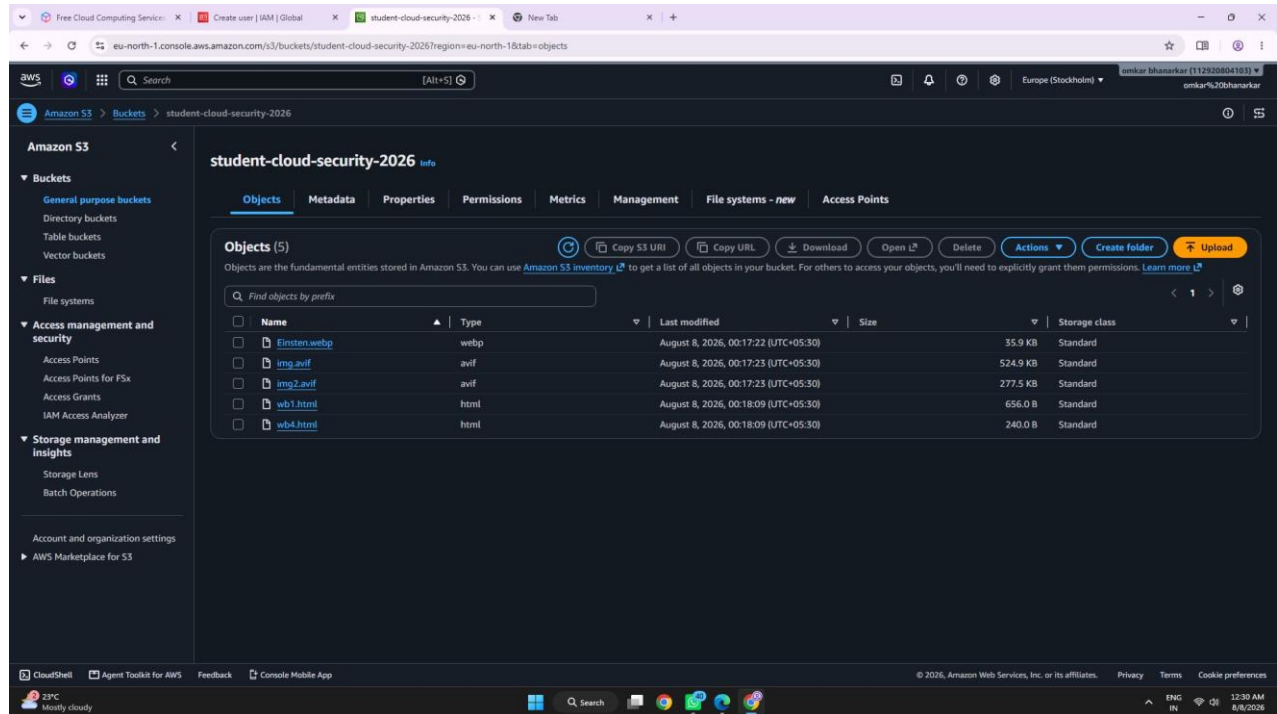
The following evidence section maps each screenshot to the corresponding implementation step. Four screenshots have been inserted from the images provided. Seven additional professional placeholders are reserved for the remaining screenshots.

Figure 1: AWS Management Console Home**Figure 1. AWS Management Console Home**

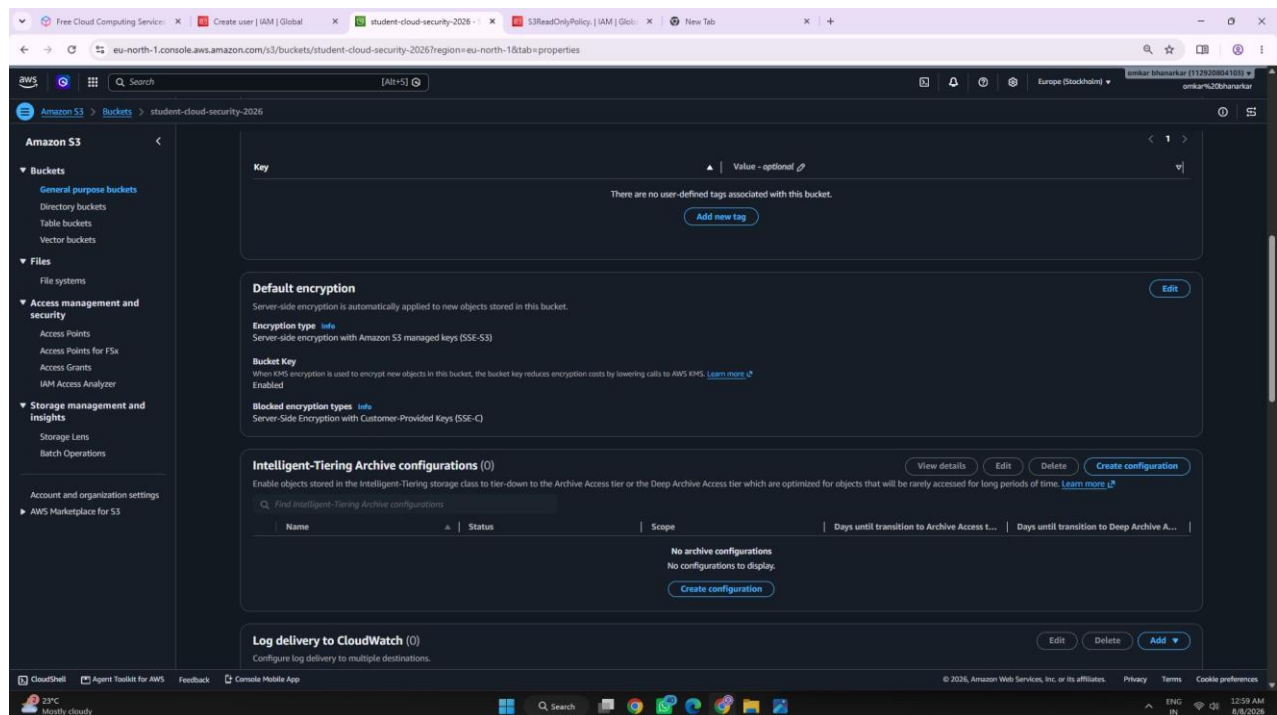
Evidence of AWS console access and available AWS services.

Figure 2: Amazon S3 Bucket List**Figure 2. Amazon S3 Bucket List**

Shows the configured S3 bucket student-cloud-security-2026 in the Europe (Stockholm) Region.

Figure 3: S3 Bucket Objects / Uploaded Files**Figure 3. S3 Bucket Objects / Uploaded Files**

Shows five uploaded objects stored in the student-cloud-security-2026 bucket.

Figure 4: S3 Default Encryption Settings**Figure 4. S3 Default Encryption Settings**

Shows server-side encryption with Amazon S3 managed keys (SSE-S3) and Bucket Key enabled.



Figure 6: IAM Policy — S3ReadOnlyPolicy



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Figure 7: IAM Policy Attached to student-user

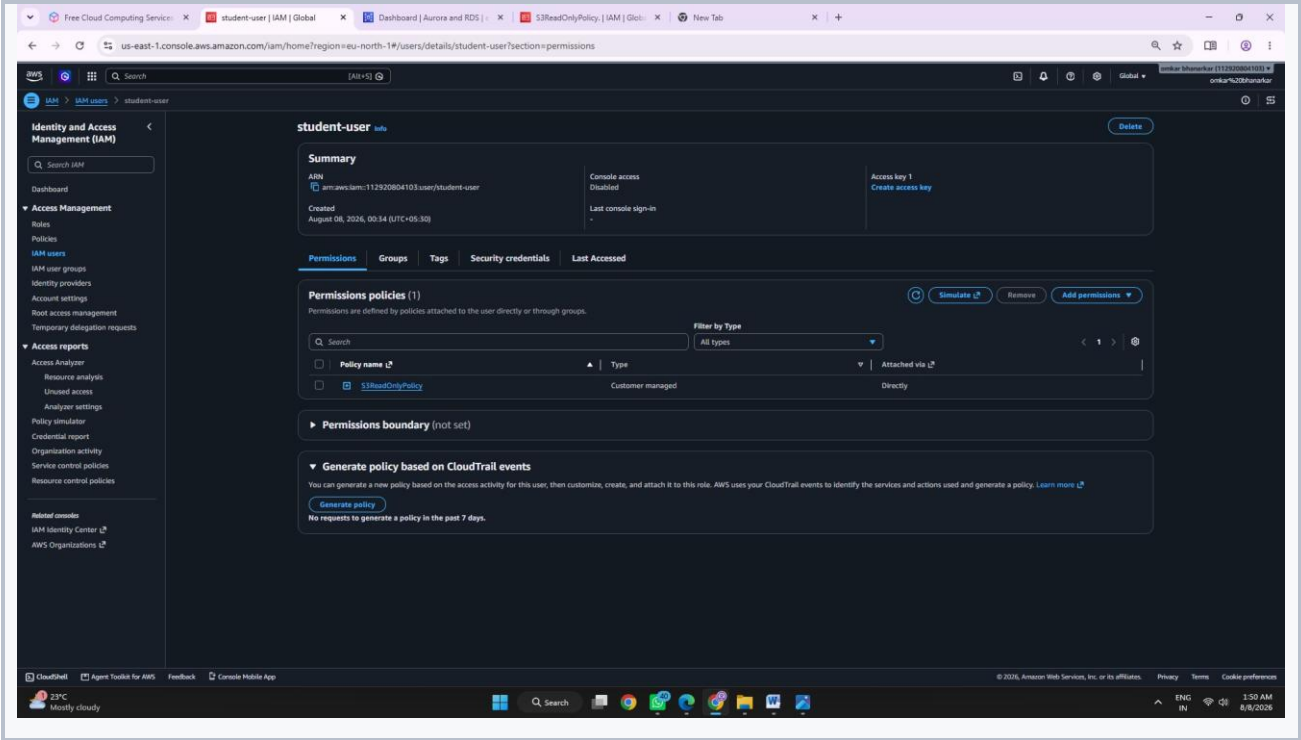


Figure 7. IAM Policy Attached to student-user — screenshot to be inserted

Insert the screenshot showing S3ReadOnlyPolicy attached directly to student-user.

Figure 8: S3 Block Public Access Settings

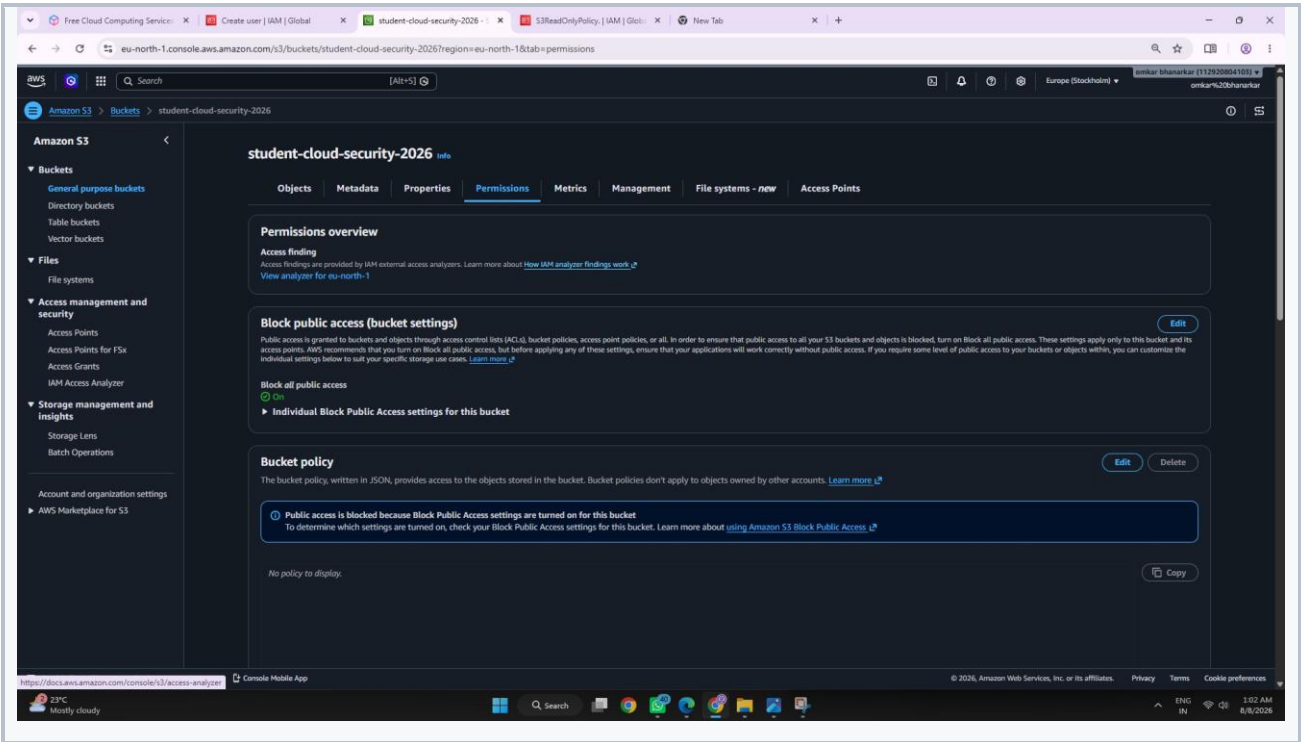


Figure 8. S3 Block Public Access Settings — screenshot to be inserted

Insert the screenshot showing Block all public access enabled for the bucket.

Figure 9: S3 Permissions / Bucket Security

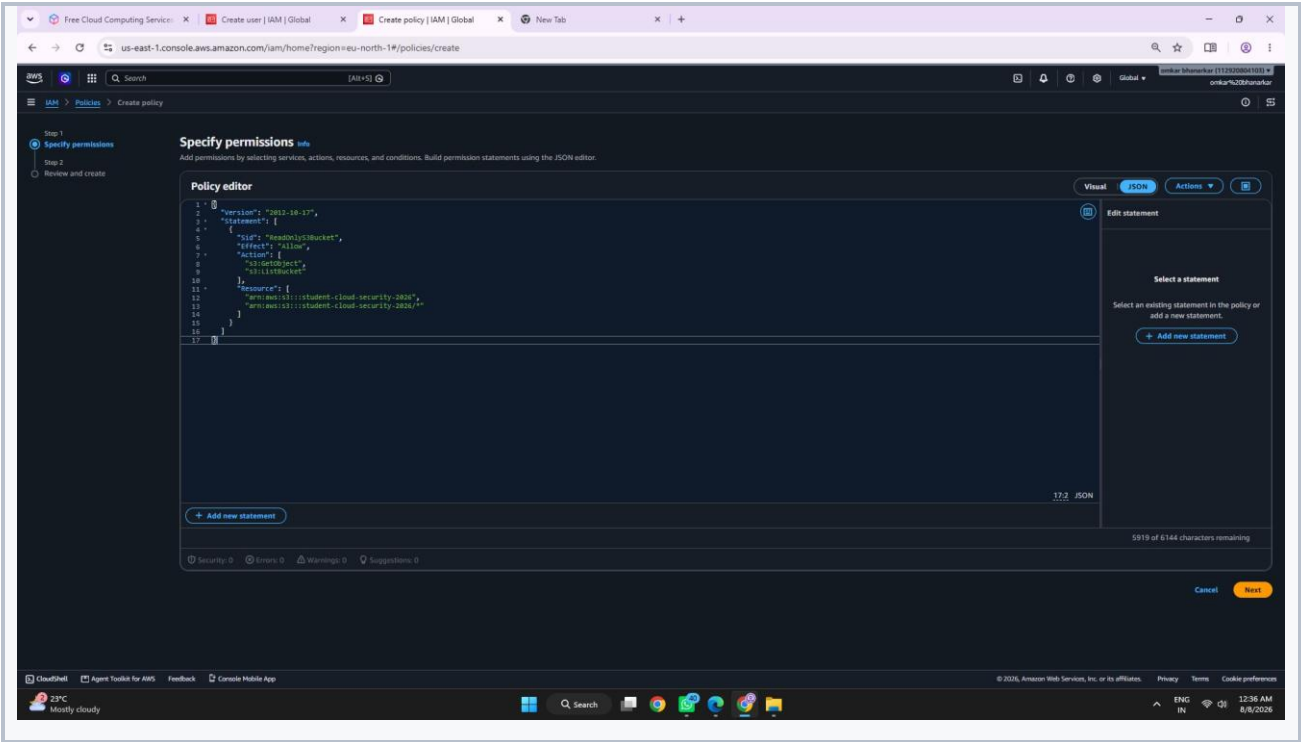


Figure 9. S3 Permissions / Bucket Security — screenshot to be inserted

Insert the screenshot showing the bucket Permissions tab and related access controls.

Figure 10: S3 Object Details

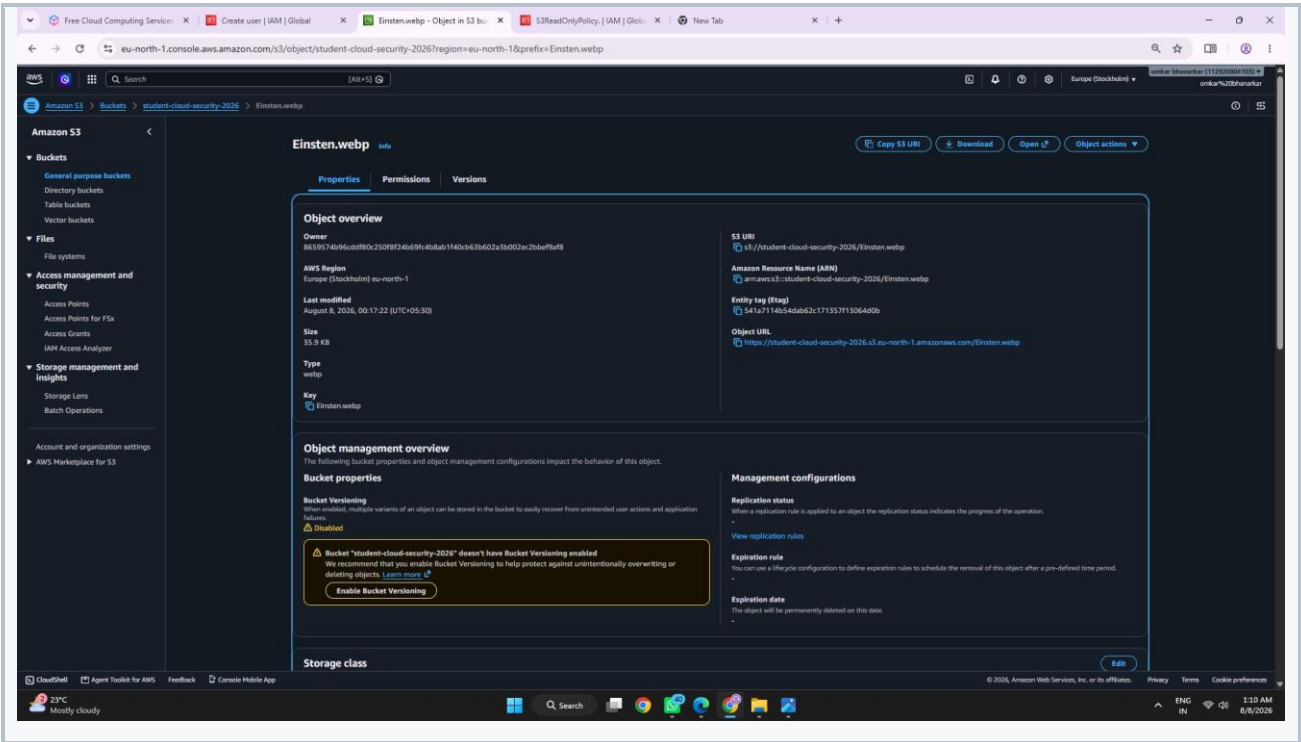


Figure 10. S3 Object Details — screenshot to be inserted

Insert the screenshot showing details of one uploaded object, if required by the practical.

Figure 11: Final Configuration / Verification

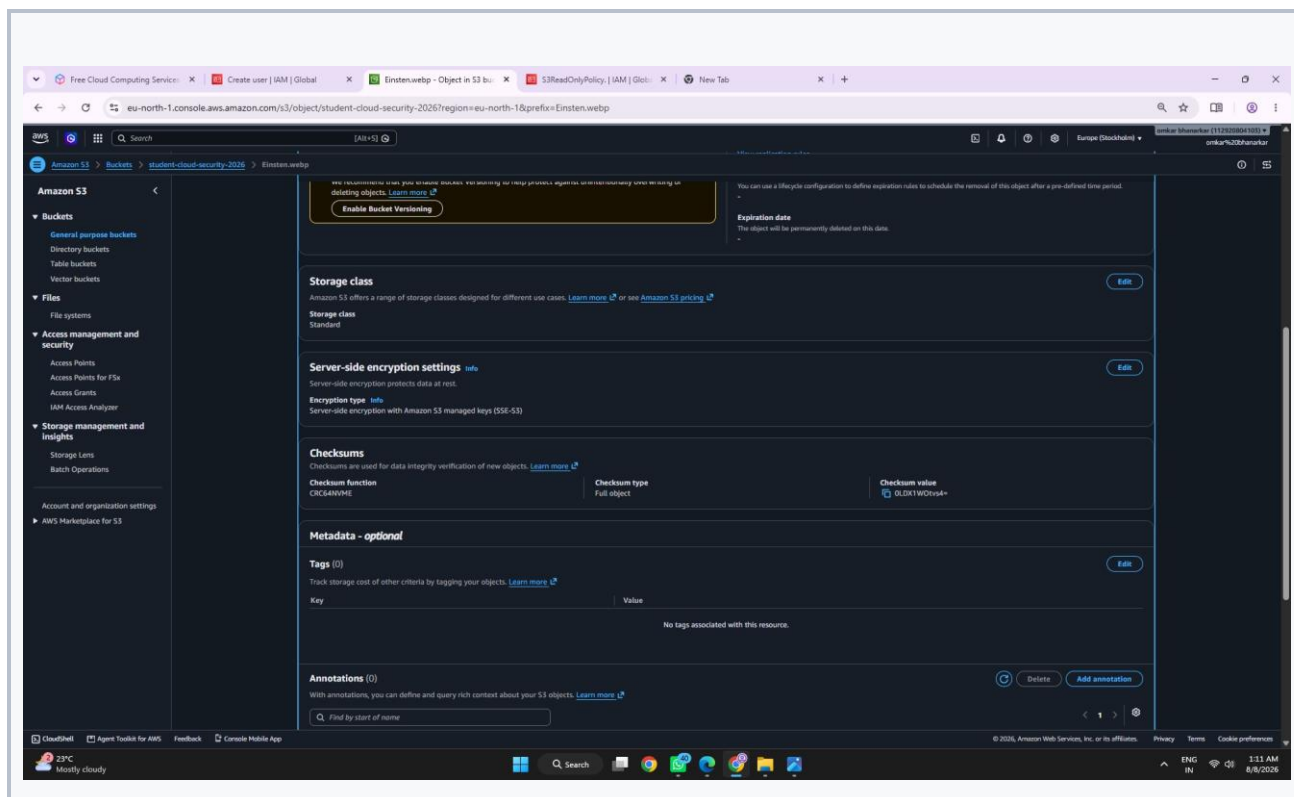


Figure 11. Final Configuration / Verification — screenshot to be inserted

Insert the final screenshot demonstrating the completed AWS security configuration.

11. Security Controls Summary

Control	Implemented Configuration	Status
Identity and Access Management	student-user with S3ReadOnlyPolicy	Implemented
Secure Object Storage	Amazon S3 bucket with controlled access	Implemented
Public Access Protection	Block public access should remain enabled	Verify / Evidence
Encryption at Rest	SSE-S3 using Amazon S3 managed keys	Implemented
Sample Data	Five objects uploaded to the bucket	Implemented

12. Result

The AWS cloud-security practical was implemented using AWS IAM and Amazon S3. The S3 bucket student-cloud-security-2026 contains uploaded sample objects and has default server-side encryption configured with SSE-S3. The student-user identity has a custom S3 read-only policy attached. The remaining evidence placeholders in this report can be replaced with the corresponding console screenshots before final submission.

13. Conclusion

This practical demonstrates the basic implementation of cloud security controls on AWS. IAM provides identity and authorization management, Amazon S3 provides secure object storage, and SSE-S3 provides encryption at rest. Together, these controls reduce the risk of unauthorized access and help protect data stored in the cloud. The practical also demonstrates the importance of documenting configuration evidence through screenshots.

14. References

- AWS Management Console — Amazon Web Services.
- Amazon S3 — Object Storage Service.
- AWS Identity and Access Management (IAM).
- AWS S3 Server-Side Encryption (SSE-S3) documentation.

END OF REPORT